

CODEX Neural – Use Case Scenarios

Derived from the current CODEX Neural implementation and written in a report-ready format.

UC-01 – Register

Actor	Research User
Precondition	No existing account.
Main Flow	Enter email/password → validate input → POST /api/register → create user record.
Postcondition	Account is created in SQLite.

UC-02 – Login

Actor	Research User
Precondition	Registered account exists.
Main Flow	Enter credentials → POST /api/login → authenticate → store authenticated user in the React session.
Postcondition	Authenticated workspace opens.

UC-03 – Reset Password

Actor	Research User
Precondition	Registered email exists.
Main Flow	Enter email → request OTP → OTP sent through SMTP → verify OTP → submit new password.
Postcondition	Password is reset; user can sign in.

UC-04 – Research Topic

Actor	Research User
Precondition	Logged in; Nexus mode selected.
Main Flow	Enter topic → FastAPI /api/research → search agent uses web search → first returned URL is scraped → writer synthesizes
Postcondition	Research report is returned and displayed.

UC-05 – Analyze PDF

Actor	Research User
Precondition	Readable PDF selected.
Main Flow	Upload PDF → extract text and sections → split into chunks → generate embeddings → store in ChromaDB.
Postcondition	Document ID is returned and the PDF becomes queryable.

UC-06 – Ask Document Question

Actor	Research User
Precondition	PDF has been indexed.

Main Flow	Enter question → MMR retrieves relevant chunks filtered by document ID → LLM answers using the retrieved PDF context.
Postcondition	Answer and page references are displayed.

UC-07 – Admin Dashboard

Actor	Admin
Precondition	Logged-in email matches the configured administrator.
Main Flow	Open dashboard → backend verifies admin → query users and login_activity → return counts and records.
Postcondition	Admin sees registered users and authentication activity.

UC-08 – Manage Users / Login Activity

Actor	Admin
Precondition	Admin dashboard is accessible.
Main Flow	Select a record → confirm removal → backend verifies admin → delete record → refresh dashboard.
Postcondition	Selected record is removed.